



“The jet fighter is one of the great icons of the second half of the twentieth century, a symbol of achievement, of technical excellence, of ultimate modernity, of latent military power.”

IVAN RENDALL

FROM HIS BOOK *ROLLING THUNDER*

HI-TECH WARFARE

A NEW GENERATION OF MILITARY AIRCRAFT USING ADVANCED ELECTRONICS AND PRECISION WEAPONS HAS MADE AIR POWER MORE EFFECTIVE THAN EVER

IN JUNE 1982, THE SYRIAN AIR FORCE sent up its MiG-21 and MiG-23 fighters to engage the Israeli Air Force over the Bekaa Valley in Lebanon. In three days of large-scale air combat, the Syrians lost 50 aircraft shot down by Israel's American-supplied F-15 and F-16 fighters. Not a single Israeli fighter was lost in air-to-air combat. Syria's ground air-defense system was almost equally ineffectual, downing only one piloted Israeli aircraft. As usual the skill and aggression of Israeli pilots had much to do with the outcome, but this could not account for the unprecedented superiority achieved. The Israeli triumph was the sign of revolutionary progress in aviation technology and electronics. This was a revolution that was set to continue through to the 1990s, when the second-order state of Iraq would stand almost in the same relation to US air power as Sudanese tribesmen had to the Maxim gun a century earlier.

The era of the modern fighter can be dated to the introduction of the US Navy's F-14 Tomcat and the USAF's F-15 Eagle in the mid-1970s. These no-expense-spared aircraft were marvels of technological innovation. They had enormously powerful engines, generating around twice the thrust available to any previous fighters, and they were packed with radar and computer equipment that helped the aircraft to new levels of performance in both flying and fighting.

Arguably the first modern fighter, the F-14 had a variable-geometry wing, the degree of sweep automatically adjusted by computer for optimum performance in all situations.

The two crew sat in a large glass bubble giving



INSIDE AN F-14 TOMCAT

The view from the two-man cockpit of a US Navy F-14 Tomcat is all-around and unimpeded. The variable-geometry wing is here in the swept-back position that the onboard computers automatically select for supersonic flight.

TORNADO GR.1

The Panavia Tornado, produced as a joint venture by Britain, Germany, and Italy, is a Mach 2 variable-geometry-wing fighter that first flew in 1974. This GR.1 version is an interdiction/strike aircraft.

